THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect square? 1

(a) 46 **(b)** 51 **(c)** 49 **(d)** 55

2. The value of 12^2 is: 1

(a) 132 **(b)** 144 **(c)** 24 **(d)** 156

3. $\sqrt{169}$ is equal to: 1

(a) 14 **(b)** 13 **(c)** 15 **(d)** 12

4. The value of 4^3 is: 1

(a) 12 **(b)** 48 **(c)** 68 **(d)** 64

5. $\sqrt[3]{64}$ is: 1

(a) 6 **(b)** 3 **(c)** 4 **(d)** 5

6. The sum of the first 5 odd natural numbers is: 1

(a) 24 **(b)** 30 **(c)** 25 **(d)** 10

7. Which of the following is a perfect square? 1

(a) 70 **(b)** 64 **(c)** 66 **(d)** 61

8. The value of 14^2 is: 1

(a) 182 **(b)** 196 **(c)** 28 **(d)** 210

9. $\sqrt{289}$ is equal to: 1

(a) 19 **(b)** 17 **(c)** 18 **(d)** 16

10. The value of 5^3 is: 1

(a) 15 **(b)** 125 **(c)** 75 **(d)** 130

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of 27^2. 2

12. Is 225 a perfect square? Give a reason. 2

13. Find the value of 13^3. 2

14. Find the value of 33^2. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find $\sqrt{196}$ by prime factorisation. 3

16. Find $\sqrt[3]{512}$ by prime factorisation. 3

17. Write 36 as the sum of the first 6 odd natural numbers (show the sum pattern). 3

OR

Find $\sqrt{324}$ by prime factorisation. 3

18. Find $\sqrt[3]{729}$ by prime factorisation. 3

OR

Write 49 as the sum of the first 7 odd natural numbers (show the sum pattern). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the smallest number by which 180 must be multiplied so that the product is a perfect square. Also find the square root of the product. 5

OR

Find the smallest number by which 108 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient. 5

20. Find the smallest number by which 252 must be multiplied so that the product is a perfect square. Also find the square root of the product. 5

OR

Find the smallest number by which 500 must be divided so that the quotient is a perfect cube. Also find the cube root of the quotient. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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